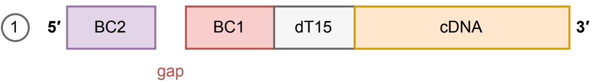
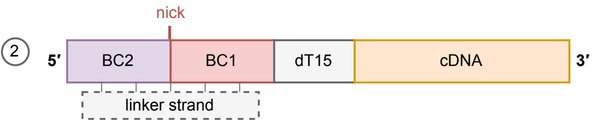
(a) Splint ligation — a ligase JOINS two ends

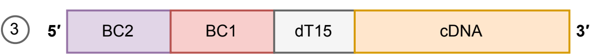
Rounds 2 and 3 of split-pool barcoding, and only those: barcode 1 arrives on the round-1 RT primer. A barcode is a sequence the protocol chose, so it must be delivered as a physical oligo.



Two separate molecules. The incoming barcode oligo sits beside the growing strand and is not attached to it. Both ends already exist as DNA.



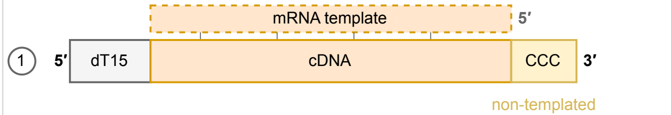
The linker base-pairs across the junction and holds the two ends abutted. It is a jig, not a template: T4 DNA ligase seals the nick, and the linker itself is never copied into anything.



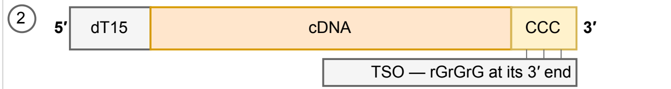
One continuous strand, one nucleotide bond longer than the two pieces were. The linker is displaced by a blocking strand and discarded. **Nothing was synthesized.**

(b) Template switching — a polymerase COPIES an oligo

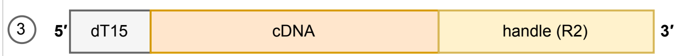
How the second PCR handle is added. After lysis the barcoded cDNA is captured on streptavidin beads and given a fresh reverse-transcriptase reaction, with the TSO supplied free in the mix.



One molecule, and the enzyme is still on it. Reverse transcriptase has copied the mRNA, reached its 5′ end, and added two or three C residues that no template asked for.



The TSO base-pairs with those C residues while the enzyme is still engaged. No melting, no re-annealing, no second primer: this is the same reaction, still running.



The enzyme switches template and carries on, so the handle is *written into* the cDNA rather than joined to it. The TSO stays a template and is not part of the product. **New sequence was synthesized.**

Ligase. Joins two existing ends. The oligo that guides it is a jig and is thrown away; what ends up in the product is the *barcode* oligo, which was already DNA. SPLiT-seq calls the jig a linker strand, scifi-RNA-seq a bridge oligonucleotide; splint is the general name for what it does.

Polymerase. Extends one end by copying. The oligo that guides it (the TSO) is a *template* and is also thrown away. What ends up in the product is a fresh copy of it.

Both appear in the same protocol, and the reason is what each end looks like when its step begins. A barcode is an arbitrary sequence the protocol picked, so no template for it exists anywhere in the cell and it has to be delivered as an oligo and joined on. The second handle sits at the far end of the cDNA, where reverse transcriptase stopped. Brettner et al.'s bead step is not run for the handle alone: the in-cell reaction is hobbled by formaldehyde crosslinking, so a second reverse-transcriptase pass is needed anyway, with the barcoded cDNA priming its own elongation. The TSO rides along on a step the protocol already had to run.

A 3′ library still template-switches, and this is the point most easily got wrong. Template switching is not a 5′-chemistry feature: it is how nearly every short-read single-cell library gets a PCR handle onto its second end, and a molecule with a handle at one end only cannot be amplified. What 3′ and 5′ actually name is *which oligo carries the cell barcode*. In a 3′ library (10x 3′, and the split-pool chemistry here) the barcode rides on the oligo-dT that primes reverse transcription, and the TSO is supplied free and carries nothing but the handle. In a 5′ library the two swap: the TSO is on the gel bead and carries the barcode and UMI, and the oligo-dT is the free primer. Both template-switch. Only one of them puts the barcode at the transcript's 5′ end, which is what makes the 5′ kit able to read a guide by its scaffold.